

AgriDirect — AI-Driven Direct-to-Buyer Agricultural Marketplace

Smart India Hackathon Problem Statement 26033

AgriDirect is a high-converting, mobile-responsive Web App frontend engineered to eliminate mandi middleman exploitation (saving 12–18% margins), provide real-time ML crop price and demand predictions, ensure 100% financial security via Smart Escrow, and reduce farm logistics costs by up to ~30% through multi-stop route aggregation.

Architectural Overview & Integration Alignment

1. RESTful Express/Node API Standard Payloads

- **Crop Registry:** `GET /api/v1/crops` & `POST /api/v1/crops`
- **Escrow Contracts:** `POST /api/v1/orders/escrow-lock`
- **Fleet Dispatch & Route Optimization:** `POST /api/v1/logistics/optimize-route`

2. MongoDB Document Model Alignment

Components consume standard document models structured as:

```
{
  "_id": "crop_65a8e101f31a2b001c",
  "farmerId": "fpo_nashik_084",
  "fpoName": "Sahyadri Farmers Producer Co. Ltd",
  "cropName": "Nashik Red Onion",
  "category": "Vegetables",
  "quantityKg": 8500,
  "minPriceExpectationPerKg": 28.5,
  "mandiPricePerKg": 31.0,
  "location": {
    "address": "Pimpalgaon Baswant, Niphad Taluka, Nashik, MH 422209",
    "lat": 20.1754,
    "lng": 73.9872
  },
  "qualityGrade": "A+",
  "status": "active",
  "isVerifiedFPO": true,
  "escrowEnabled": true,
  "timestamp": "2026-08-28T09:30:00.000Z"
}
```

3. Python Machine Learning Service Predictions

The UI natively renders predictions generated by backend ML services:

- `predictedPricePerKg` (Linear/XGBoost mandi price forecasting)
- `demandIndex` (0–100 B2B demand score categorized as High, Medium, Low)
- `recommendedRoutePolyline` & `distanceSavedPercent` (Dijkstra / OR-Tools multi-stop route pooling)

4. Offline-First Resilience

- Automatically detects offline state and routes newly logged produce batches to `localStorage` queue.
- Shows live top banner status: "`Saved locally – auto-syncing when online`".
- Features a built-in "**Simulate Offline / Go Online**" toggle in the navigation bar to test offline resilience and instant auto-synchronization.



Key Modules

1. FPO / Farmer Management Dashboard (`/fpo-dashboard`):

- Harvest Batch Logging with Geo-coordinate pinning & 1-click regional presets.
- Mandi Price vs ML Forecast Line Chart (Recharts) with disintermediation gain metrics (+14.2%).
- Farmer's Registered Batches Table with verified FPO badge and status tracking.

2. B2B Buyer Marketplace (`/marketplace`):

- Live Produce Catalog with category tabs, search query, quality grading, and price filters.
- Demand Forecasting Widget displaying real-time category procurement gauges.
- Smart Order Drawer & Escrow Breakdown Modal:
 - `Base Crop Cost` + `Consolidated Route Freight` - `Middleman Commision Saved` = `Net Escrow Locked`.

3. Route Optimization & Freight Dispatch (`/logistics`):

- Interactive SVG GIS Radar Map Viewer with live GPS telemetry, vehicle tracking, and multi-stop waypoints.
- Transit Efficiency Metrics: Distance Saved (-78 km), Freight Saved (₹4,900 / 29.8%), CO₂ Emissions Reduced (142.6 kg).
- Pickup Waypoint schedule with status checkpoints (Completed, In Progress, Scheduled).



Getting Started Locally

Prerequisites

- Node.js (v18 or newer recommended)
- npm or yarn

Installation

```
# 1. Clone repository & install dependencies
npm install
```

```
# 2. Start Vite development server
npm run dev

# 3. Build production bundle
npm run build
```

Open <http://localhost:3000> in your browser.